

Open book exam. Submit your answers to gradescope.

In your explanations and justifications, write complete and grammatically correct sentences.

Please do not ask questions during the exam.

3. Suppose we have a single resource and 8 requests to be scheduled, given by the list of tuples $L = [(0, 4), (9, 10), (7, 10), (3, 9), (8, 12), (5, 6), (7, 13), (7, 9)]$, where each tuple (s, f) gives the start time s and finish time f of each request.
- (a) Use the example to illustrate the scheduling of the requests such that no requests overlap and that the number of scheduled requests is maximal.
 - (b) Explain why the greedy rule you applied leads to an optimal solution. Illustrate your explanations using the example.

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Answer:

3. (a) The sorted requests are $L = [(0, 4), (5, 6), (3, 9), (7, 9), (9, 10), (7, 10), (8, 12), (7, 13)]$.
- i. accepting request $(0, 4)$, keeping request $(5, 6)$, removing overlapping request $(3, 9)$, keeping requests $(7, 9), (9, 10), (7, 10), (8, 12), (7, 13), (5, 6), (7, 9), (9, 10), (7, 10), (8, 12), (7, 13)$
 - ii. accepting request $(7, 9)$, keeping request $(9, 10)$, removing overlapping requests $(7, 10), (8, 12), (7, 13)$
 - iii. accepting request $(9, 10)$
- The accepted requests are $[(0, 4), (5, 6), (7, 9), (9, 10)]$.
- (b) The greedy algorithm stays ahead, implying that in any alternative optimal set, the finish time in any alternative set of requests should be either the same or later than the finish time of the accepted requests computed by the greedy rule. This ensures that the number of accepted requests in any alternative optimal set cannot be larger than what the greedy rule computes.